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RESEARCH ARTICLE

A Tri-Level Semantic and LLM-Assisted Methodology for Systematic Literature Reviews (TriSem-LLM)

TASSIA CARVALHO^{1,2}, SARA GARCÊS^{3,4}, AND SANDRA JARDIM¹

¹Smart Cities Research Center, Polytechnic University of Tomar, 2300-313 Tomar, Portugal

²University of Beira Interior, 6201-001 Covilhã, Portugal

³Polytechnic University of Tomar, 2300-313 Tomar, Portugal

⁴Instituto Terra e Memória—Geosciences Center, 6120-750 Mação, Portugal

Corresponding author: Sandra Jardim (sandra.jardim@ipt.pt)

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ABSTRACT Systematic reviews of the literature increasingly involve large, multilingual, and heterogeneous corpora, making traditional keyword-based approaches inefficient and potentially prone to the exclusion of relevant studies. This paper proposes TriSem-LLM, a multi-stage screening methodology that integrates keyword filtering, multilingual semantic similarity at the abstract and full-text levels, and large language model (LLM)-assisted synthesis within a unified and fully traceable workflow. Unlike conventional screening tools, the proposed approach avoids early exclusion and evaluates all candidate articles through multiple complementary criteria, enabling the recovery of studies that may not be captured by lexical matching alone. The methodology is designed to prioritize recall while maintaining transparency, modularity, and reproducibility across all processing stages. The approach is evaluated through a structured validation procedure, including a manually annotated subset and an extended inspection of excluded articles. Within the validation subset (44 articles classified as relevant and 44 as non-relevant), full agreement between automated and manual decisions was observed in this specific context. An additional inspection of excluded articles identified two cases of false negatives, indicating that relevant studies may still be missed despite the multi-stage screening process. These results should be interpreted as an initial indication of the internal consistency and practical feasibility of the proposed methodology rather than as a definitive estimate of global performance. In general, the findings suggest that TriSem-LLM can support transparent and reproducible screening workflows in systematic reviews, particularly in multilingual and interdisciplinary contexts where semantic variability and terminology differences pose significant challenges.

INDEX TERMS Systematic literature review, semantic screening, document prioritization, multilingual embeddings, large language models, semantic similarity.

I. INTRODUCTION

The systematic review of the literature is a well-established methodological approach aimed at rigorously identifying, evaluating, and synthesizing the available scientific evidence related to a specific research question [1]. During the past decade, its relevance has increased substantially, driven by the rapid growth of academic production in multiple fields of

knowledge. This expansion, together with the linguistic diversity of bibliographic repositories and the heterogeneity of scientific terminology, has made it increasingly challenging to ensure that systematic reviews achieve comprehensive coverage while maintaining transparency and reproducibility [2]. As a result, the initial screening stage has become a critical bottleneck in the systematic review process, particularly when dealing with large-scale and multilingual corpora.

Among the available frameworks, the PRISMA method (*Preferred Reporting Items for Systematic Reviews and*

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Meta-Analyses) has become one of the principal international references for conducting and reporting systematic reviews, with the objective of improving clarity and reducing bias [3]. The most recent update, PRISMA 2020, introduced a revised 27-item checklist, updated flow diagrams, and specific recommendations for abstract reporting, further strengthening standards for transparency and methodological reporting [3]. In addition, the PRISMA 2020 Explanation and Elaboration document provides detailed guidance on the interpretation and application of each item, including the definition of eligibility criteria and the design of the flowchart [4]. Despite these advances, the practical implementation of the PRISMA principles remains largely manual, particularly during the initial stages of screening and eligibility evaluation, which can become time consuming and operationally demanding when dealing with large and heterogeneous datasets [5].

In response to these challenges, recent research has increasingly explored the integration of computational and machine learning techniques to support systematic review workflows. Previous studies have investigated automated or semi-automated approaches for the screening, prioritization, and management of evidence from the literature, highlighting the potential of machine learning models to improve efficiency while preserving methodological rigor [6], [7], [8], [9], [10]. More recently, transformer-based language models and sentence embedding techniques have demonstrated promising performance in semantic similarity assessment and abstract screening tasks [11], [12], [13], [14]. These developments suggest that semantic representations can partially overcome the limitations of purely keyword-based strategies, particularly in interdisciplinary contexts where relevant concepts may be expressed using diverse vocabularies.

Parallel to these advances, large language models (LLMs) have begun to emerge as auxiliary tools in systematic review processes, supporting tasks such as document screening, data extraction, and preliminary synthesis [15], [16], [17], [18]. Although recent studies indicate that LLMs can assist human reviewers when combined with appropriate validation mechanisms, concerns remain about reliability, bias propagation, and the transparency of automated decision making. Consequently, current consensus emphasizes the role of LLMs as assistive tools within human-in-the-loop frameworks, rather than as autonomous decision makers.

To provide a structured overview of existing approaches, prior work can be broadly categorized into three main groups: keyword-based and rule-based screening methods, which rely on lexical matching and predefined criteria; machine learning and active learning approaches, which prioritize documents based on iterative human feedback; and embedding-based semantic similarity methods, which leverage vector representations to capture contextual relationships between texts. Although these approaches have demonstrated improvements in efficiency and recall under specific conditions, they typically address isolated components of the screening process and do not provide a unified and fully traceable workflow.

Despite these advances, several limitations remain in existing approaches to automated or semi-automated screening in systematic reviews. First, many workflows still rely on early keyword-based exclusion strategies, which may prematurely discard relevant studies due to variations in terminology or language. Second, the integration of full-text semantic analysis remains limited despite its potential to capture deeper contextual relevance beyond abstracts. Third, current systems often lack transparent and traceable decision mechanisms, making it difficult to justify inclusion and exclusion outcomes in a reproducible manner.

To address these limitations, this study proposes a hybrid methodology named *TriSem-LLM*, designed to support the early stages of systematic literature reviews through the integration of semantic similarity techniques based on Transformer models, multilingual analysis, and a transparent multi-criteria decision logic. Unlike traditional workflows that rely on early keyword-based exclusion, the proposed approach evaluates all retrieved articles using complementary lexical and semantic criteria, avoiding premature filtering.

The semantic encoding of the research question is performed in three languages—English, Spanish, and French—allowing for a wider linguistic and cultural coverage during the selection of the article. In addition, open-source language models are employed to assist in exploratory synthesis while ensuring traceability of all intermediate decisions. This combination of multilingual semantic representation, full-text evaluation, and explicit decision traceability positions *TriSem-LLM* as a reproducible and interpretable tool within the landscape of existing screening approaches.

The main contributions of this work are as follows:

- A reproducible and modular pipeline (*TriSem-LLM*) that integrates established natural language processing techniques, including semantic embeddings, cosine similarity, and LLM-assisted synthesis, into a unified workflow tailored for systematic literature reviews.
- A multi-stage screening strategy that combines keyword filtering, abstract-level semantic similarity, and full-text semantic analysis, reducing reliance on early keyword-based exclusion while prioritizing recall.
- A multilingual semantic analysis approach that aims to improve coverage across heterogeneous and multilingual corpora.
- A transparent and traceable decision framework aligned with the PRISMA 2020 guidelines, allowing the reproducibility and auditability of screening decisions.

It is important to note that this work does not propose a novel NLP algorithm; rather, its contribution lies in the structured integration and practical operationalization of established techniques into a coherent and scalable methodology for systematic review automation.

II. RELATED WORK

Recent advances in natural language processing have significantly influenced the development of automated and semi-automated approaches for systematic literature review

screening. Existing work can be broadly grouped into three main categories.

First, keyword- and rule-based approaches rely on predefined inclusion and exclusion criteria, often based on lexical matching. Although these methods are simple and transparent, they are limited in their ability to capture semantic variability across documents, particularly in interdisciplinary and multilingual contexts.

Second, machine learning and active learning approaches aim to reduce the screening workload by iteratively prioritizing relevant documents. Previous studies have demonstrated that active learning strategies can identify the majority relevant publications while screening only a fraction of the dataset [19]. These approaches have been further developed in tools such as ASReview [14], which have shown strong performance in reducing manual effort while maintaining high recall. However, such methods typically require continuous human interaction and may lack interpretability in their decision-making processes.

Third, embedding-based and transformer-based methods leverage semantic representations to assess document similarity. Transformer-based language models such as BERT [20] and sentence embedding techniques such as Sentence-BERT [21] have demonstrated improved performance over traditional representations in text classification and semantic similarity tasks. These approaches enable the capture of contextual relationships between documents, overcoming limitations associated with bag-of-words models.

Recent studies have also explored the application of NLP and machine learning pipelines to automate title and abstract selection [22], [23], [24]. These approaches have reported strong classification performance and significant workload reduction, although they often depend on labeled training data and may face challenges in generalizing across domains and research topics.

In parallel, large language models (LLMs) have been investigated as auxiliary tools in systematic review workflows, supporting tasks such as document screening, information extraction, and preliminary synthesis [25]. Although LLM-based approaches show promising capabilities, concerns remain regarding reliability, bias, and reproducibility, reinforcing the need for transparent and verifiable screening methodologies.

Recent studies have further expanded the application of advanced NLP techniques to complex multimodal and domain-specific tasks. These include approaches to align vision-language models [26], conversational question answering in resource-constrained settings [27], and ontology-enhanced analysis of medical text [28]. Such works highlight the growing diversity and sophistication of the methods available for semantic understanding and information extraction in specialized domains.

Despite these advances, most of the existing methods focus on isolated components of the screening process rather than providing a unified and fully traceable workflow. In contrast, the proposed TriSem-LLM methodology integrates keyword

filtering, semantic similarity at both abstract and full-text levels, and LLM-assisted synthesis within a single, transparent pipeline aligned with PRISMA 2020.

III. THEORETICAL FRAMEWORK

Systematic reviews are distinguished by the adoption of explicit and replicable procedures that ensure a rigorous synthesis of the available knowledge [1], [2]. The PRISMA model, widely recognized as a reference standard, structures the process into successive stages that include record identification, duplicate removal, title and abstract selection, eligibility evaluation, and synthesis of included studies [3]. However, it is important to emphasize that PRISMA is primarily a reporting guideline rather than a methodological standard, and its misuse for protocol design or quality assessment has been frequently observed in the literature [6]. Although these stages have proven relevant for promoting transparency and reliability, the manual execution of each step involves significant constraints, particularly in terms of time, consistency, and scalability [5].

With the exponential growth of scientific production and the multiplication of multidisciplinary repositories, it is becoming increasingly challenging to maintain reviews that are up to date, comprehensive, and methodologically robust. Several studies have pointed out that the manual execution of screening and analysis processes can not only compromise timeliness but also increase the risk of biases arising from fatigue or from terminological heterogeneity between sources [2], [5]. In this context, there has been a consolidation of a research body focused on exploring the potential of AI-assisted systematic reviews. Recent scoping reviews have shown that automation techniques are being applied in multiple stages of the workflow, with most advances concentrated on the screening of records, although their real world adoption remains modest and context-dependent [8]. In addition, a rapid scoping review identified more than 100 studies employing machine learning methods at different stages of the systematic review workflow, from search and selection to data extraction and bias assessment, while highlighting barriers such as usability, reproducibility and limited large scale adoption [9].

In addition to these scoping reviews, previous contributions have also proposed model-based approaches to systematic reviews, in which the process is formally represented through structured models to improve clarity, repeatability, and integration of results [7]. Such frameworks highlight the importance of explicitly modeling the review workflow, providing a conceptual foundation for subsequent advances that incorporate computational and AI-driven components. This model oriented perspective is extended in the present study, which formalizes the review workflow not only conceptually, but also computationally, consolidating semantic and generative components into a transparent pipeline.

Methodologies have emerged that seek to reconcile the normative rigor of PRISMA with the use of computational tools capable of accelerating critical stages such as document

screening, duplicate removal, and preliminary evidence extraction. However, it is important to acknowledge that full automation of these stages raises significant concerns about decision traceability, result verifiability, and the risk of perpetuating biases embedded in the models used [5].

The most recent advances in language models based on *Transformers* have substantially expanded the range of options available to support systematic reviews. Technologies such as Sentence-BERT have been admired for their ability to generate semantic embeddings that capture deep contextual relationships between descriptors and document contents, overcoming the limitations associated with the literal matching of keywords [11], [13]. These dense vector representations have shown potential in the automatic prioritization of records and in reducing the volume of articles subjected to manual evaluation, thus contributing to greater efficiency and process traceability [12]. Recent comparative studies further confirm the superiority of Transformer-based models for abstract selection in systematic reviews, consistently outperforming conventional ML algorithms in accuracy and recall [10]. The present work expands this line of research by adopting multilingual Transformer embeddings to capture the terminological diversity between disciplines.

However, Large Language Models (LLMs) have shown increasing potential to support different stages of systematic reviews, particularly in preliminary document analysis, assisted descriptor generation, and semantic identification of similarities between texts. A concrete example is the use of models based on Sentence Transformers to classify large sets of articles according to their semantic proximity to the research question, substantially reducing the number of documents requiring manual screening [11]. Recent evaluations highlight both the potential and limitations of using LLMs such as GPT-4 in systematic reviews. Although under certain conditions their performance can approach human levels, concerns remain about reliability, prompt dependence, and bias [15]. These findings reinforce the importance of hybrid frameworks such as TriSem-LLM, which integrates LLMs under structured and transparent criteria, ensuring human oversight and alignment with PRISMA principles. Although these applications offer efficiency gains, concerns remain about the opacity of generative processes, variability in the quality of responses, and the need for careful validation by human researchers.

Beyond individual case studies, recent syntheses show that LLMs are already being adopted throughout the review workflow, from database search and abstract screening to evidence extraction and synthesis, highlighting their role as emerging tools in the production of scientific reviews [16]. This trend underscores the timeliness of the TriSem-LLM pipeline, which aligns with this broader movement by the operationalization of LLM-assisted processes in a structured and transparent manner.

In this regard, several academic initiatives have sought to develop hybrid methodologies that combine the benefits of semantic representation and generative artificial intelligence

with mechanisms of human oversight, ensuring alignment with the principles of a systematic review [17], [18]. These approaches stand out for their articulation of multilingual embeddings, lexical heuristics, and manual verification processes, configuring pipelines that aim to ensure scalability, traceability, and methodological quality simultaneously.

The present work is situated within this conceptual framework and proposes an innovative solution that combines these components in a structured manner applied to the PRISMA methodology, whose detailed description will be presented in the Methodology section. Although existing reviews [9] provide a valuable mapping of machine learning methods for isolated tasks, they also underline the fragmented and heterogeneous nature of current solutions. In contrast, TriSem-LLM contributes by offering an integrated, multilingual, and traceable architecture aligned with PRISMA, aiming to operationalize automation in a cohesive and reproducible manner.

IV. PROPOSED METHODOLOGY

This section details the new automated pipeline *TriSem-LLM*, which combines multilingual semantic embeddings, keyword heuristics, and LLM-based synthesis to support systematic reviews of the literature. The proposed methodology employs Transformer-based language models to compute semantic similarity between research questions and scientific articles, integrating the average of multilingual embeddings, the calculation of cosine similarity, and lexical matching for relevance classification. The aim is for this automated process to reflect, in the screening and preliminary selection stage, the structural principles defined by the PRISMA method, with respect to the sequential logic of the initial steps and the traceability criteria outlined in its flowcharts. Some critical concerns have been raised about the use of rigid exclusion criteria in the initial stages of systematic reviews. In this article, unlike previous approaches, TriSem-LLM: (i) completely avoids early exclusion filters based on keywords; (ii) applies trilingual semantic encoding to the research question; (iii) segments and analyzes the full text with vector embeddings; and (iv) uses open generative models to synthesize evidence from the recommended articles. The proposal aims to mirror the PRISMA structure, ensuring traceability and multilingual coverage in document screening.

TriSem-LLM: *Multilingual Transformer-based semantic pipeline with LLM-driven synthesis for systematic reviews.*

In this research, follow the PRISMA 2020 guidelines, which replace the 2009 version and detail new recommendations, including the use of updated flowcharts and detailed descriptions of the screening process [3], [4]. Furthermore, the PRISMA-S extension [5] was adopted to ensure transparent and reproducible reporting of literature search, covering aspects such as database selection, search strategies, limits, filters, updates, and de-duplication procedures.

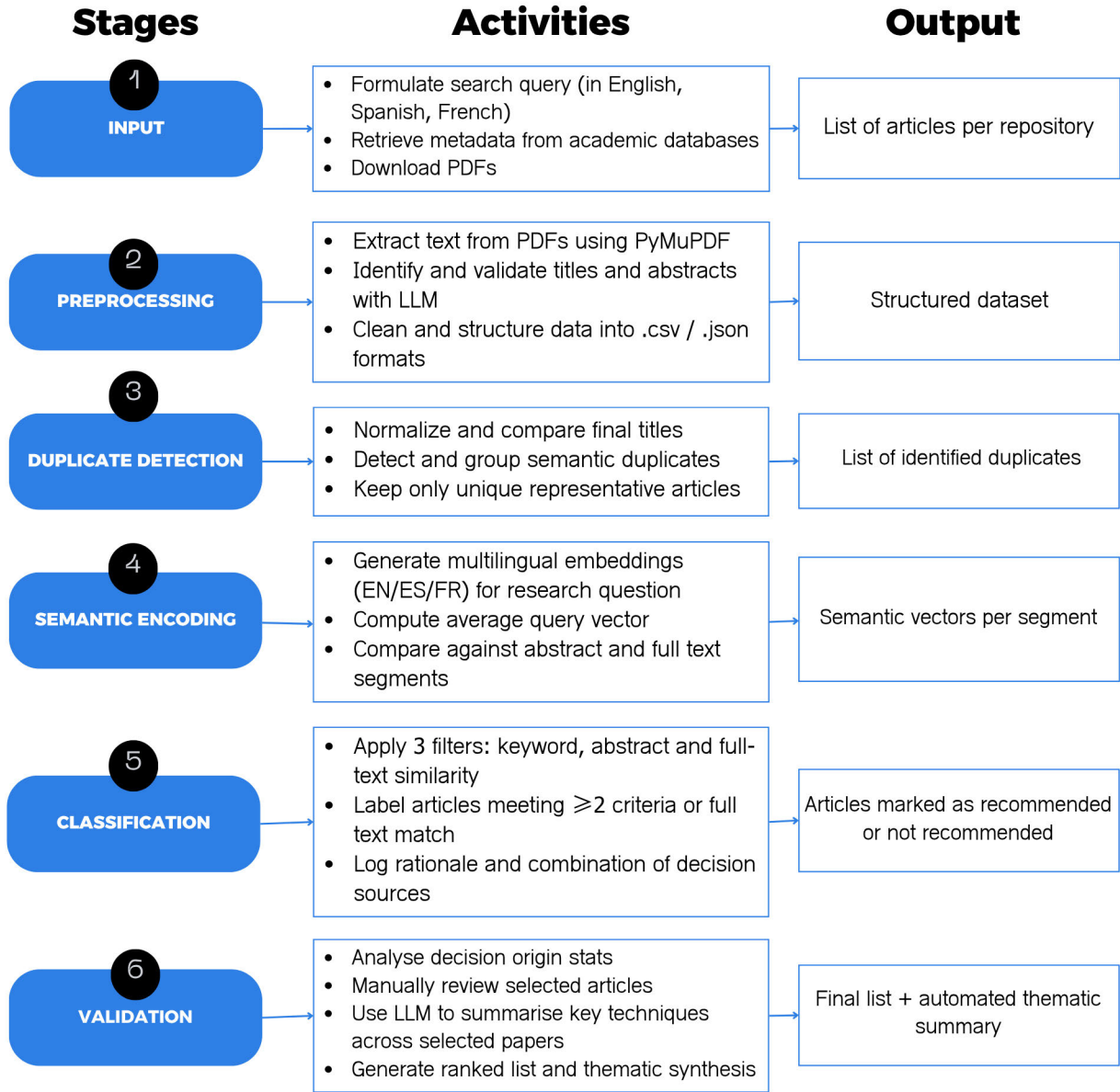


FIGURE 1. Workflow of the automated TriSem-LLM pipeline. The diagram presents the six main stages of the proposed methodology, from data input to final validation. Each phase includes specific activities, such as metadata extraction and verification, duplicate detection, semantic encoding, multi-criteria classification, and validation through statistical analysis, manual review, and LLM-assisted synthesis.

To complement the visual representation in Fig. 1, the main semantic similarity computations underlying the workflow are formally defined below.

The core of the methodology is based on the pretrained model *all-mpnet-base-v2*, from the *SentenceTransformers* library, which generates dense semantic embeddings for both abstracts and full texts. To ensure linguistic diversity and maximize recall, the research question was translated into English, Spanish and French, and the generated embeddings were combined into a single vector average. Subsequently, for all the documents, cosine similarity was calculated with this averaged vector, both from the abstracts and the full texts. In parallel, lexical verification was also

applied by keywords. These three analyzes were performed independently and without early exclusion, allowing the identification of relevance using different criteria. In this pipeline, Transformer models act as semantic encoders, while similarity scores and lexical matches constitute the core of the automated screening logic.

Formally, let q denote the encoded research query and d_i a candidate document. The abstract-level similarity score is computed as:

$$S_{\text{abs}}(d_i) = \cos(\mathbf{e}(q), \mathbf{e}(\text{abstract}_i)) \quad (1)$$

where $\mathbf{e}(\cdot)$ represents the embedding function.

For full-text analysis, each document is segmented into k chunks and the similarity is computed as:

$$S_{\text{full}}(d_i) = \max_{j=1,\dots,k} \cos(\mathbf{e}(q), \mathbf{e}(c_{i,j})) \quad (2)$$

where $c_{i,j}$ denotes the j -th text segment of document d_i .

TABLE 1. Summary of documents retrieved and downloaded by source.

Source	Retrieved	Downloaded	Notes / Query Description
SCOPUS	161	62	General multilingual query
Web of Science	67	40	Same structure as SCOPUS
ScienceDirect	545	472	Query 1: "image processing" AND "rock art"; Query 2: "digital imaging" AND "rock painting/engraving"; Query 3: broader variation (491 results)
MDPI	25	24	Mostly unrelated content
IEEE Xplore	121	117	Focus on image/signal processing
Springer	49	27	Manually filtered for relevance
Taylor & Francis	207	57	Filtered subset used

The final decision score is obtained through a multi-criteria combination of keyword matching, abstract similarity, and full-text similarity:

$$S_{\text{final}}(d_i) = f(S_{\text{keyword}}, S_{\text{abs}}, S_{\text{full}}) \quad (3)$$

The decision function $f(\cdot)$ represents a rule-based aggregation strategy that combines lexical and semantic criteria without early exclusion.

Additionally, a comparative structure was maintained with the traditional workflow based on early exclusion by keywords, whose comparative analysis is presented in a later section, with the aim of assessing the methodological impact of this choice on the coverage and relevance of the retained articles.

A. DATA SOURCES AND PREPROCESSING

The dataset was compiled from several leading academic databases, including SCOPUS, Web of Science, ScienceDirect, MDPI, IEEE Xplore, Springer, and Taylor & Francis. Each retrieved article record included metadata such as title, abstract, and a link to the full text PDF when available in open access.

The search queries were primarily formulated using the following structure:

TITLE-ABS-KEY(("digital imaging" OR "image processing" OR "computer vision" OR "artificial intelligence" OR "machine learning" OR "deep learning") AND ("rock art" OR "prehistoric art" OR "rock painting" OR "rock engraving" OR "petroglyphs" OR "pictograms"))

The searches were restricted to articles in English, Spanish, or French. Only open-access documents were downloaded due to access constraints; therefore, pay-per-view articles were not included in the dataset. In total, 799 documents were downloaded. Table 1 summarizes the number of documents retrieved and successfully downloaded from each source. Multiple queries were used in Science Direct to ensure broader coverage.

Following data collection, a preprocessing stage was applied to ensure consistency and quality of the dataset. Duplicate records across different databases were identified and removed. Textual data, including titles and abstracts, were normalized to reduce formatting inconsistencies. In addition, full-text documents were segmented into smaller units to enable fine-grained semantic similarity analysis.

The resulting dataset was used as input to the automated screening pipeline and subsequently for the manual validation procedures described in Section V-C. This structured preprocessing ensures the reproducibility and traceability of the data preparation process.

B. TEXT EXTRACTION FROM PDF AND DATA STRUCTURING

The scientific articles in PDF format were processed using the PyMuPDF library to extract the full text and associated metadata. A customized pipeline was implemented to detect and collect the file name, the article title, the abstract, the type of publication, and the full content.

Title extraction was performed through a detailed analysis of the first page of each document, identifying text blocks with the largest average font size and upper vertical position. Only blocks with a minimum of 6 words, appropriate font size, and no occurrence of generic or recurrent terms (e.g., journal, abstract, cover, among others) were considered. Blocks composed exclusively of ignored expressions (e.g., conference names or generic expressions) were also excluded. The first valid candidate, ordered by vertical position, was assumed to be the main title of the article.

Abstract identification was carried out using regular multilingual expressions applied to the first three pages of the document, capable of recognizing typical headers such as "Abstract", "Resumen", or "Résumé", including variations with irregular spacing or special characters. When it was not possible to identify a structured abstract, complementary heuristics were applied: (i) detection of large blocks containing typical introductory phrases (e.g., this paper, este artículo, cet article); (ii) extraction of text immediately preceding the keyword section; (iii) selection of the first substantial block on the first page; or, as a last resort, (iv) extraction of content located before the formal introduction of the article (1. Introduction or I. INTRODUCTION). Each abstract was annotated with the corresponding extraction strategy.

Publication identification was performed on the basis of regular multilingual patterns that searched for explicit mentions of journals, conferences, or proceedings. Headers such as journal, revista, conference, proceedings, as well as expressions such as published in, publicado en, and publié dans were considered. Detection was applied to the full text of the article, and the first valid occurrence was assumed to be the source of the publication.

The full text of each article was extracted and saved in a plain text file (.txt), preserving the structure of the original repository. In parallel, the main metadata — file name, title, abstract, abstract source (extraction strategy label), publication name, full text file path and repository

subfolder — were collected into a structured dataset using the Pandas library. This dataset was exported in.csv format, facilitating process reproducibility and subsequent semantic analysis.

To support manual verification of the data and ensure extraction quality, an additional visual export step was implemented. This step consisted of automatic generation of an HTML table, where file names and full text paths were converted to dynamic hyperlinks. Each link pointed directly to the original PDF file or to the respective extracted text file, opening them in the default browser of the system. This visualization allowed for quick and interactive inspection of metadata — with particular focus on validating titles and publication names —, facilitating the identification of errors such as the incorrect extraction of journal names as main titles.

To further improve metadata quality, an additional validation stage was introduced using a Large Language Model (LLM) through the Together.ai API.¹ Detection of potentially invalid titles was based on a verification function that flagged as suspicious those that were too short, generic, or matched recurrent terms, including a curated list of examples to ignore.

In each flagged case, the first 3,000 characters (or the first page) of the full text were submitted to the *mistralai/Mixtral-8 × 7B-Instruct-v0.1* model, configured with a temperature of 0.3 to ensure consistent output. The *prompt* was specifically designed to instruct the model to return only the likely article title, without any additional explanation.

The generated title was stored in a new column (*title_llm*), while also retaining the heuristically extracted original title. In the next stage, a final column (*title_final*) was created, prioritizing the LLM corrected title whenever available. This consolidation was essential for de-duplication, following PRISMA guidelines for the removal of redundant records with documented traceability, grouping entries with equivalent titles, and generating unique file pairs for manual review.

The process also included fault tolerance mechanisms, such as verifying the existence of text files and error handling for temporary API limitations (such as codes 429 and 503), ensuring that exceptions did not interrupt the overall execution flow.

C. MODEL CONFIGURATION AND MULTILINGUAL EMBEDDING STRATEGY

To perform a semantic similarity analysis between the scientific articles and the previously defined research question, the pretrained model *all-mpnet-base-v2*, available in the *SentenceTransformers* library, was used. This model is based on the Transformer architecture and is recognized for its strong ability to capture semantic relationships in multilingual contexts. Execution was configured adaptively: whenever available, processing was delegated to the GPU to maximize efficiency in embedding generation.

The choice of *all-mpnet-base-v2* was motivated by its consistent performance in semantic textual similarity tasks, particularly in systematic literature review scenarios. In a comparative evaluation involving four pretrained models and more than 6,000 articles, Zhang and Lencioni demonstrated that this model achieved high average cosine similarity scores (0.77–0.80), reliably ranked relevant documents among the top retrieved results [11].

The research question was translated and encoded into three languages — English, Spanish, and French — in order to maximize semantic coverage and reduce linguistic bias. The three generated embeddings were combined by vector averaging, producing a single representative vector of the scientific query. This strategy reinforces the language agnostic nature of *TriSem-LLM*, allowing it to identify semantically relevant articles in multilingual corpora.

1) TEXT PREPROCESSING AND ENCODING STRATEGY

The abstracts were automatically filled in from the extracted metadata, with missing values handled. For the full texts, the .txt files resulting from prior PDF extraction were loaded. Both abstracts and full texts were subjected to the same vector encoding process to be semantically analyzed.

To mitigate the context limitations of the Transformer models (512 tokens), the full text was segmented into adaptive blocks, compatible with the model's token limit. Each segment was encoded separately, and the maximum cosine similarity between any block and the query vector was considered to be the semantic relevance of the document.

2) ORDERED APPLICATION OF RELEVANCE FILTERS

Automated screening was organized into three main filters, applied independently and sequentially, with transparent decision records in separate columns:

- 1) **Keyword filter:** The first criterion assessed the simultaneous presence of technical and thematic terms in the abstract, with multilingual coverage (English, Spanish, and French). These included technical terms related to digital image processing, machine learning, and computer vision, as well as thematic terms associated with rock art and archaeological heritage. An article was considered positive under this filter if the abstract contained at least one occurrence from each group.
- 2) **Abstract similarity filter:** Cosine similarity was calculated between abstract embeddings and the multilingual query vector. The relevance threshold for the semantic similarity score was empirically set at 0.60. This value was chosen after a preliminary analysis of a small, manually validated subset of articles, which indicated that scores at or above this threshold consistently corresponded to documents of high relevance to the research topic. Although some articles with a lower score (e.g., between 0.50 and 0.60) were also found to be relevant, a threshold of 0.60 was found to offer an optimal balance between precision and

¹<https://api.together.xyz/v1/chat/completions>

recall, effectively filtering out the majority of irrelevant documents while retaining a comprehensive set for subsequent analysis. This value was selected based on empirical observations during preliminary exploration of the corpus, where lower thresholds admitted a substantial number of weakly related articles, while higher thresholds increased the risk of excluding semantically relevant contributions expressed using heterogeneous terminology. It is important to note that this threshold is a parameter that can be adjusted depending on the specific domain and the desired balance between sensitivity and specificity. Specific details of the similarity calculation are provided in the following subsection.

- 3) **Full text similarity filter:** Following the segmented strategy, each article was divided into text blocks, and each segment was encoded for comparison with the query vector. The highest similarity was taken as representative of the article. The same 0.60 threshold was applied.

The cosine similarity threshold used for filtering was determined through an initial calibration process based on a small manually inspected subset of approximately 25 articles. This subset was selected to include a representative mix of articles with varying degrees of semantic similarity to the research topic, enabling exploratory analysis of inclusion and exclusion behavior across different threshold values. Importantly, this calibration subset was disjoint from the validation subset used in Section V-C and was not used for performance evaluation.

During this preliminary analysis, lower threshold values resulted in a substantial increase in candidate articles, introducing a higher proportion of clearly non-relevant items, while higher thresholds led to the exclusion of potentially relevant studies. These observations highlight the sensitivity of the threshold parameter in balancing recall and precision in similarity-based screening.

Based on this trade-off, a threshold of 0.60 was selected as a balanced operating point that prioritizes recall while maintaining a manageable set sizes candidate. Although no complete quantitative sensitivity analysis was conducted, the extended validation presented in Section V-C, including the identification of false negatives, provides empirical support that the selected threshold maintains high recall while limiting the risk of premature exclusion. Alternative application domains, corpora characteristics, or research objectives may require different threshold values.

The decision not to apply a sequential exclusion workflow, which would immediately discard articles failing the first criterion, was based on the need to reduce the risk of omitting potentially relevant works that, while not explicitly mentioning the keywords in the abstract, might contain substantive content aligned with the research question. Thus, all articles were subjected to each filter independently, ensuring that strong semantic signals identified in the full text or abstract could recover records that would otherwise

be discarded. This logic aimed to prioritize the sensitivity (*recall*) of the process, acknowledging that the author's terminology may vary significantly depending on the journal, the disciplinary area, or the original language. Additionally, in order to assess the impact of this choice, an alternative version of the pipeline with immediate sequential exclusion after the lexical filter was implemented, as prescribed in the PRISMA guidelines. The comparison between the two workflows — with and without early exclusion — is presented in Figure 2, highlighting the differences in the retrieval of relevant articles.

3) SEMANTIC SIMILARITY CALCULATION

Semantic similarity was measured using the cosine similarity metric, applied to the vector embeddings of the abstract and the full text relative to the averaged query vector. For each article, two independent scores were calculated: one score associated with the abstract and another with the full body text. This parallel evaluation was designed to maximize the robustness of the screening, ensuring that contextual evidence present only in one of the sources could still contribute to the general relevance.

To ensure that the semantic evaluation of the full text was not compromised by the limitation of context window of the Transformer model, each article body was pre-segmented into blocks compatible with the 512-token limit. Each block was individually encoded and the highest similarity between any segment and the research question was taken as the representative measure of the relevance of the document. This strategy allowed automatic truncation to be bypassed and enabled a more comprehensive analysis of the article content. The approach proved effective in addressing the limitations of Transformer context windows, allowing for a broader semantic evaluation. Future work may explore segmentations guided by document structure (e.g., separating sections such as introduction, methods, results) and the adoption of models with expanded context windows, further strengthening the coverage and granularity of the screening.

Additionally, it was observed that some articles with thematically relevant content yielded cosine similarity scores slightly below the defined threshold (0.60), clustering in the 0.50 range. Although this interval is, in principle, indicative of weak semantic association in a general purpose model, subsequent qualitative analysis revealed that such articles contained valid contributions to the research question. Therefore, it is proposed that in highly specialized domains or with a non-standardized vocabulary, a pre-selection zone with a reduced threshold (e.g., 0.50–0.59) should be considered, provided it is complemented by manual validation.

4) MULTI-CRITERIA COMBINATION IN THE VERSION WITHOUT EARLY EXCLUSION

The results of each filter were recorded in separate columns, ensuring the complete traceability of the criteria satisfied by each article. Based on these data, two pipeline variants were developed: one version with immediate sequential exclusion

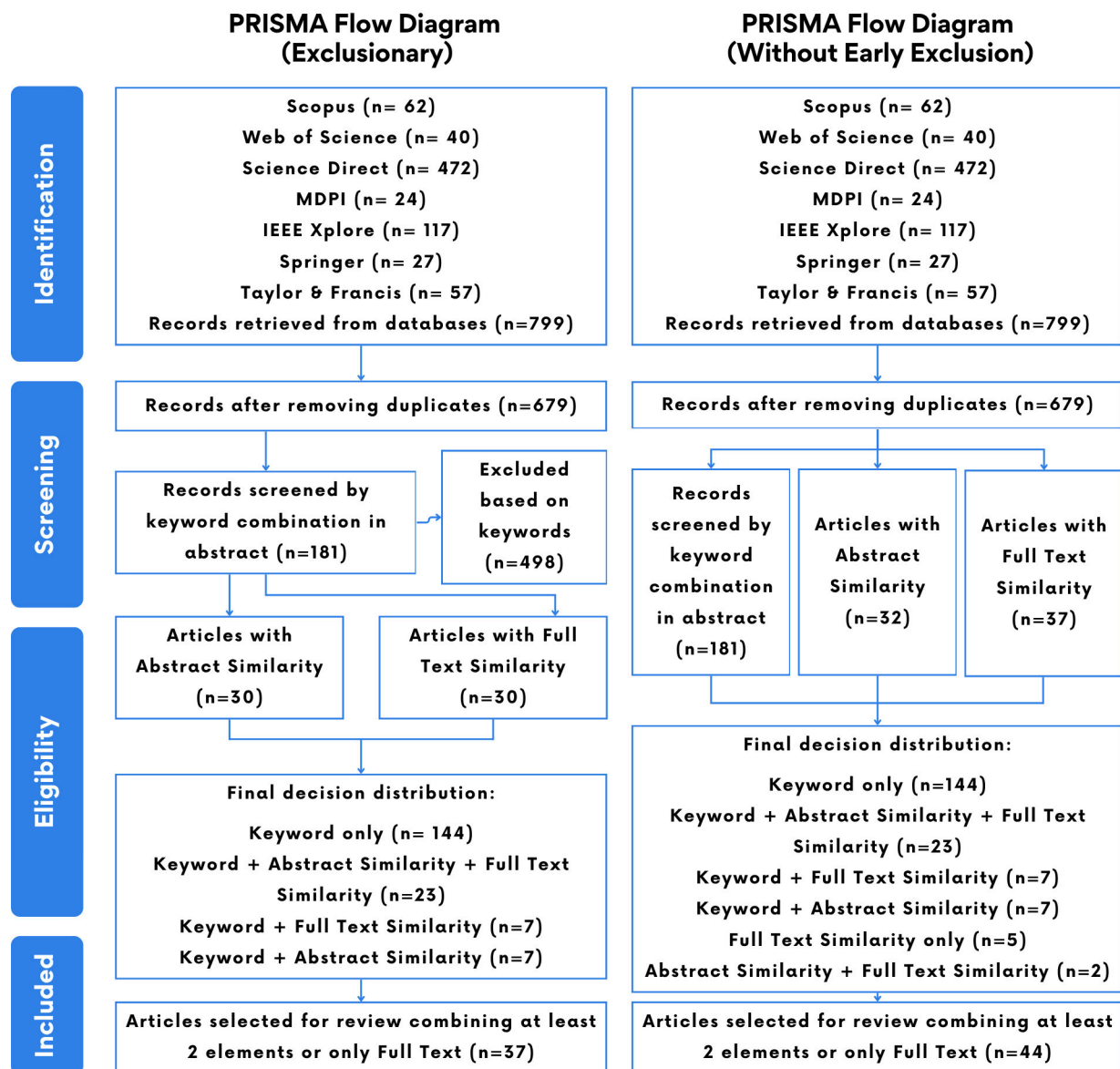


FIGURE 2. Comparison between the traditional PRISMA workflow (with early keyword-based exclusion) and the workflow adopted in this study (without initial exclusion and with multi-stage semantic analysis). The diagram highlights the differences between early keyword screening in the abstract and the proposed approach, which retains all records until semantic similarity evaluation with the full text.

after the lexical filter, aligned with the traditional PRISMA workflow, and another with multi-criteria evaluation, without early exclusion, as proposed in this study. In the multi-criteria version, the data were combined into a consolidated decision column, which indicated, for each article, all sources of evidence justifying its recommendation for manual review. This approach allowed for the retrieval of potentially relevant articles that could have been prematurely discarded if only a single filter had been applied in isolation. The comparison between the two approaches is presented in Figure 2, which highlights the impact of early exclusion on the sensitivity of the screening process.

It is important to note that, for recommendation and final consideration purposes, only the version without early exclusion was used, allowing all articles to be screened based on the three filters in parallel.

The adopted recommendation logic defined that any article meeting the combination of at least two criteria, or solely the criterion of full text similarity, would be marked as *recommended for analysis*:

- Keyword + Abstract + Full Text
- Keyword + Full Text
- Keyword + Abstract
- Abstract + Full Text

- Full Text

In addition, a column was created to record the highest similarity score between abstract and full text, allowing articles to be ranked by degree of semantic alignment. This ranking was used to generate a prioritized list of the 30 articles with the greatest potential relevance to guide manual review.

5) EXPORT AND ORGANIZATION OF RESULTS

The final subset of articles recommended for review was exported in CSV format, containing all calculated indicators, similarity metrics, keyword matches, and the consolidated decision source. This structure ensured full traceability of decisions and allowed complete replication of the process across different environments.

To facilitate manual review, an automated procedure was implemented to physically organize the PDF files according to the classification assigned to each article. Separate directories were created based on the combination of criteria met (for example, *Keyword*, *Full Text*, *Keyword_Abstract_Full Text*), and the PDFs were copied into their respective folders. This hierarchical organization was designed to optimize qualitative verification segmented by the profile of evidence identified.

6) COMPARISON OF PRISMA STRATEGIES: EARLY EXCLUSION VS. FULL ANALYSIS

In traditional PRISMA-aligned workflows, systematic reviews are typically organized around progressive screening of documents, beginning with the analysis of titles and abstracts. In this approach, articles that do not contain keywords or apparent relevance are excluded early, before any deeper content analysis. This strategy aims to reduce the volume of data to be processed; however, it carries the risk of eliminating relevant publications with atypical vocabulary or non-standardized terminology.

As an alternative, the present study adopted an approach without early exclusion, opting to retain all initially collected articles, even those that did not directly match the defined keywords. Instead of discarding them, these articles were later evaluated using multilingual semantic embeddings and contextual similarity analysis, taking into account the full content of the texts. This methodological decision aims to maximize *recall* and avoid premature exclusion of contributions relevant to the research question.

Figure 2 illustrates, side by side, the two approaches compared: on the left, the traditional workflow with early exclusion based on keywords; on the right, the alternative workflow adopted in this study, which applies multiple analytical criteria before any exclusion.

This hybrid strategy demonstrated a broader coverage and allowed the identification of articles that, although not meeting the initial lexical criteria, contained clear semantic evidence of relevance. Such results support the feasibility of AI-assisted systematic reviews, particularly in interdisciplinary domains with varied vocabulary.

7) PRIORITIZATION AND READING RECOMMENDATIONS

A ranked list of 30 articles with the highest overall similarity between abstract and full text was generated. This prioritized list, exported in tabular formats and accompanied by the main relevance metrics, constituted the structured basis for an in-depth qualitative review and discussion of the results.

Considering the diversity and heterogeneity of the corpus analyzed, it is recommended that the initial qualitative analysis preferably focus on articles that meet multiple selection criteria, such as “Keyword + Abstract + Full Text”, “Keyword + Full Text”, or “Keyword + Abstract”. This decision aims to increase precision without compromising thematic coverage, and can be adjusted according to the complexity of the domain under study.

8) FINAL CONSIDERATIONS ON AUTOMATED SCREENING

The adoption of this multi-criteria process made it possible to combine the robustness of semantic representation with the interpretative simplicity of lexical matches. This combined approach ensured greater flexibility and sensitivity in the initial screening while preserving the transparency and auditability of the results. Future evaluation of the agreement between this system and manual review will constitute an essential step in evaluating its effectiveness, including the analysis of potential false positives and false negatives.

V. RESULTS AND DISCUSSION

This section presents an exploratory analysis of the automated screening conducted by the *TriSem-LLM* pipeline, focusing on the characterization of relevance decisions and semantic similarity signals rather than formal performance evaluation. It describes the distributions of relevance decisions, the similarity scores obtained, the composition of the prioritized subset for review, and the validation assisted by large language models. The analyses reported in this section are descriptive in nature and aim to provide insight into the behavior of the proposed multi-criteria screening approach.

A. EXPLORATORY ANALYSIS OF AUTOMATED SCREENING

Of a total of 679 processed articles, 44 were classified as recommended for review according to the defined multi-criteria logic, representing approximately 6.48% of the initial corpus. Figure 3 (a) shows the division between recommended and unrecommended articles, while Figure 3 (b) presents the distribution of selected articles according to the combinations of criteria.

Analysis of combinations that led to selection revealed that most articles (37, or 5.44%) satisfied the keyword matching criterion combined with at least one semantic criterion. The remaining articles were distributed as follows: *Keyword + Abstract + Full Text* (23), *Keyword + Full Text* (7), *Keyword + Abstract* (7), only *Full Text* (5), and *Abstract + Full Text* (2). These results highlight the value of the multi-criteria model, particularly in enabling the inclusion

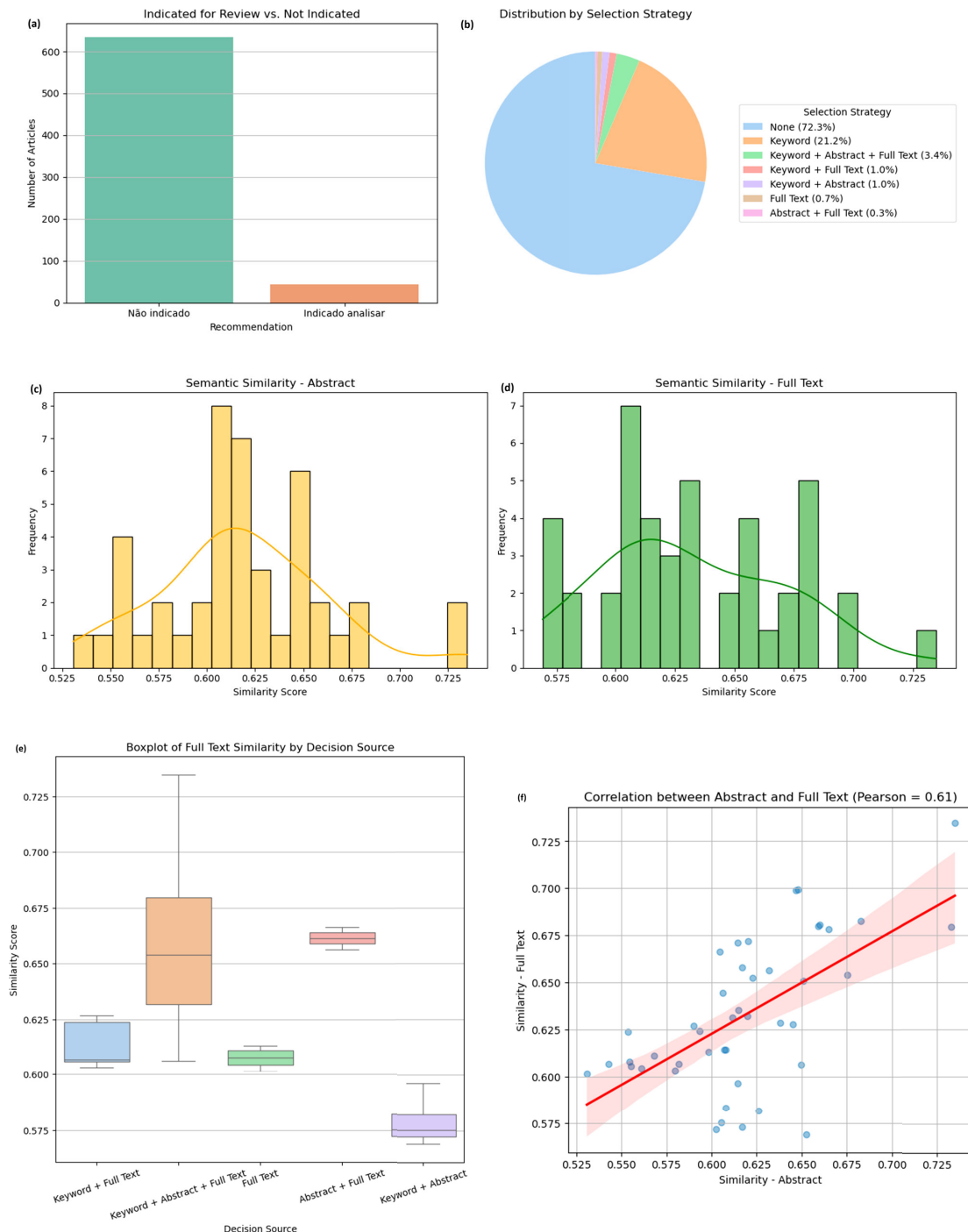


FIGURE 3. Exploratory analysis of automated semantic screening results. The figure shows: (a) proportion of recommended and non recommended articles; (b) distribution of decisions by selection strategy; (c) and (d) histograms of semantic similarity scores between the research question and the abstracts/full texts; (e) distribution of full text scores by combination of criteria. A moderate alignment is observed between abstract and full text scores (correlation of 0.61), along with a tendency toward more concentrated score distributions for articles selected by multiple criteria; and (f) correlation between the two levels of similarity.

of articles whose relevance was only identified in the full text semantic analysis.

The distribution of similarity scores revealed significant variability between the two types of representation.

Histograms (Figure 3 (c) and (d)) show that the similarity values derived from the full texts tend to be higher and slightly more uniform compared to the abstracts. A concentration of values between 0.60 and 0.68 was observed for both

cases, with a slight advantage for the full text in terms of semantic consistency. This pattern suggests that semantic analysis applied to the full text may provide a richer representation of contextual alignment with the research question, complementing the more concise and often generic nature of abstracts.

To evaluate the internal consistency of the model, the correlation between the similarity scores of the abstracts and the full texts was calculated, which yielded a Pearson coefficient of $r = 0.61$ (Figure 3 (f)). Although the correlation is moderate, the observed alignment between the two levels of analysis indicates that both contribute semantically coherent information. Nevertheless, the presence of cases with discrepancies justifies the adoption of a dual approach and complements the multi-criteria decision process.

The box plot presented in Figure 3 (e) shows the distribution of similarity scores calculated from the full text, stratified according to the combination of criteria that motivated the selection of the article. It can be observed that groups integrating multiple decision sources — particularly *Keyword + Abstract + Full Text* and *Abstract + Full Text* — exhibit higher median similarity values and more concentrated distributions, suggesting stronger semantic consistency with the research question.

Conversely, articles included solely based on partial combinations, such as *Keyword + Abstract* or only *Full Text*, display greater dispersion and lower median values. This pattern indicates the potential relevance of the complete semantic criterion as a support for automated screening, while also supporting the use of multiple sources of evidence for more robust inclusion decisions. This outcome is consistent with observations reported by Du et al., who highlighted the advantages of Transformer-based models over traditional machine learning approaches in abstract screening tasks [10].

Finally, a prioritized ranking was generated with the 30 articles that achieved the highest maximum similarity (considering the highest score between abstract and full text). This list was aimed at guiding qualitative analysis and subsequent critical validation.

B. LLM-ASSISTED EXPLORATORY SYNTHESIS

After automated selection of articles for review, a complementary exploratory analysis was conducted using large language models (LLMs). This step aimed to assess whether the articles retained by the screening could support automated evidence extraction and a preliminary synthesis of trends related to the research question. This step is not intended as a formal evaluation procedure, but rather as an exploratory mechanism to assess the feasibility of automated evidence extraction and synthesis.

Each relevant article was isolated into an individual block and paired with an optimized version of the research question, expressed in English, accompanied by strict instructions requiring: (i) the identification of image processing techniques applied to rock art; (ii) the year or period of application, when available; (iii) a concise description of the

pipeline presented in the article; and (iv) the enumeration of limitations and future perspectives mentioned by the authors. The instructions explicitly reinforced the obligation to base the response solely on the textual content provided, prohibiting external inference or use of additional knowledge not present in the articles.

The *prompts* were generated programmatically from the full text of the articles and sequentially submitted to the *Mixtral-8 × 7B-Instruct-v0.1* model via the Together.ai platform. The text generation configuration included a temperature of 0.7 and a maximum limit of 2048 *tokens*, aiming to balance creativity and consistency.

The generated responses were stored in individual files (*.txt* and *.json*), preserving the granularity of the extracted evidence. Subsequently, a consolidated synthesis file (*summary_all_blocks.txt*) was produced, aggregating the outputs obtained from all the articles analyzed. This file served as the basis for constructing the timeline of applied techniques, as well as for structuring the predominant methodological pipeline in the domain.

Additionally, a final *prompt* was designed to identify the ten most relevant articles, specifying for each: (i) the main extracted techniques; (ii) the source blocks; and (iii) a technical justification of its relevance. The final selection exhibited a degree of internal coherence and alignment with the research question, suggesting that the articles analyzed may provide a suitable basis for exploratory synthesis within the scope of this study.

This approach complemented the exploratory quantitative analysis with an AI-generated qualitative perspective, providing an additional perspective on articles with greater technical density and methodological alignment. Although it does not replace critical human reading, the LLM-assisted evaluation suggests the feasibility of integrating semantic screening and generative synthesis into a coherent pipeline, with the potential to support contemporary systematic reviews in an efficient, traceable and reproducible manner.

This partially reflects the findings of Khraisha et al., who reported that GPT-4 achieved human-like performance under specific conditions of full text screening with carefully engineered prompts, but only moderate results in most other tasks [15]. In contrast, the present study deliberately employed the open access model *Mixtral-8 × 7B-Instruct-v0.1*, prioritizing transparency, accessibility, and ease of reproducibility over reliance on proprietary solutions. This design choice ensured that the pipeline could be independently replicated by other researchers, even in contexts with infrastructure constraints, while still aligning with best practices for hybrid human–AI collaboration.

C. QUANTITATIVE EVALUATION AND MANUAL VALIDATION OF SCREENING DECISIONS

To provide an initial assessment of the behavior of the proposed screening methodology, a quantitative analysis was conducted based on a manually annotated validation subset. A confusion matrix was constructed to characterize

Rank		Filename	Similarity (Abstract)	Similarity (Full Text)	Decision Source
0	1	Table-of-contents_2015_Digital-Applications-in...	0.734814	0.734814	Keyword + Abstract + Full Text
1	2	Digital-Enhancement-Techniques-in-Prehistoric-...	0.732657	0.679473	Keyword + Abstract + Full Text
2	3	Computer-vision-methods-and-rock-art-Towards-a...	0.647912	0.699141	Keyword + Abstract + Full Text
3	4	Digital-image-analysis-based-study--recording-...	0.646761	0.698830	Keyword + Abstract + Full Text
4	5	Digital-Image-Processing-and-the-Recording-o_2...	0.682780	0.682780	Keyword + Abstract + Full Text
5	6	An-assessment-of-methods-for-the-digital-enhan...	0.659856	0.680443	Keyword + Abstract + Full Text
6	7	Evaluating-conventional-and-advanced-visible-i...	0.658998	0.679667	Keyword + Abstract + Full Text
7	8	3560-11524-1-PB.pdf	0.664939	0.678405	Keyword + Abstract + Full Text
8	9	Digital-image-enhancement-for-recording-rupest...	0.675437	0.653833	Keyword + Abstract + Full Text
9	10	Digital Image Processing Applied.pdf	0.620312	0.671677	Keyword + Abstract + Full Text
10	11	Laboratory-and-in-situ-assays-of-digital-image...	0.614331	0.671252	Keyword + Abstract + Full Text
11	12	Combining-geometrical-and-radiometrical-fea_20...	0.604418	0.666477	Abstract + Full Text
12	13	Digital-image-analysis-based-strategies-for-qu...	0.617072	0.657814	Keyword + Abstract + Full Text
13	14	The-application-of-state-of-the-art-technologi...	0.631859	0.656381	Abstract + Full Text
14	15	A-propositive-method-of-documental-registratio...	0.652537	0.568998	Keyword + Abstract
15	16	Digital-imaging-techniques-for-recording-and-a...	0.622751	0.652395	Keyword + Abstract + Full Text
16	17	Editorial_2015_Digital-Applications-in-Archaeo...	0.650746	0.650746	Keyword + Abstract + Full Text
17	18	Digital image processing in weathering damage.pdf	0.649430	0.606312	Keyword + Abstract + Full Text
18	19	Applications-of-digital-photography-in-the-st...	0.645139	0.627830	Keyword + Abstract + Full Text
19	20	A-step-further-in-rock-art-digital-enhanc_2019...	0.606400	0.644427	Keyword + Abstract + Full Text
20	21	3D-digital-documentation-and-image-enhancement...	0.638161	0.628402	Keyword + Abstract + Full Text
21	22	Colorbased-discrimination-of-color-hues-in-roc...	0.614784	0.635103	Keyword + Abstract + Full Text
22	23	Rock-art-technology-Reflectance-Transformation...	0.619877	0.632088	Keyword + Abstract + Full Text
23	24	Methodological-considerations-of-integrating-p...	0.611586	0.631378	Keyword + Abstract + Full Text
24	25	Combining-photogrammetry-and-photographic-enha...	0.590080	0.626734	Keyword + Full Text
25	26	RTI-and-the-study-of-engraved-rock-art--A-re-e...	0.626176	0.581723	Keyword + Abstract
26	27	03-Diffracting Digital Images in the Making.pdf	0.593575	0.623990	Keyword + Full Text
27	28	Modern Techniques for Ancient Art Deep Learnin...	0.553536	0.623631	Keyword + Full Text
28	29	Latest-developments-in-rock-art-recording--tow...	0.616755	0.572899	Keyword + Abstract
29	30	Digital-image-enhancement-with-DStretch---I_20...	0.614371	0.596310	Keyword + Abstract

FIGURE 4. Ranking of the 30 articles with the highest semantic similarity. The figure shows the prioritized articles based on the highest cosine similarity score, considering both abstracts and full texts. The last column identifies the combination of criteria that justified their recommendation. This ordering serves as the basis for the qualitative analysis stage, allowing us to focus on the cases with the strongest semantic alignment with the research question.

the agreement between automated screening decisions and manual assessment.

Based on this matrix, the following metrics were calculated:

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

$$F1 - score = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \quad (6)$$

$$Specificity = \frac{TN}{TN + FP} \quad (7)$$

Table 2 presents the confusion matrix derived from the validation subset, while Table 3 summarizes the corresponding performance metrics.

The inclusion of recall is particularly important in this context, as the proposed methodology is designed to minimize

TABLE 2. Confusion matrix of the automated screening results based on the manually annotated validation subset. The matrix reports the number of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), comparing the system's classification with manual assessment.

	Relevant	Not Relevant
Selected	44	0
Not Selected	0	44

the risk of excluding relevant articles during the screening process. In addition, specificity provides information on the system's ability to correctly filter out non-relevant articles.

To assess the consistency between automated screening results and human judgment, a manual validation was conducted on a stratified subset of retrieved articles. The subset consisted of 44 articles classified as relevant and 44 articles classified as non-relevant by the proposed methodology,

TABLE 3. Descriptive agreement measures derived from the confusion matrix, including precision, recall, F1 score, and specificity. These values characterize the agreement between the automated screening results and the manual assessment within the validation subset and should not be interpreted as corpus-level performance estimates.

Metric	Value
Precision	1.0
Recall	1.0
F1 score	1.0
Specificity	1.0

which were subsequently reviewed according to predefined inclusion and exclusion criteria. Manual validation was conducted by a single reviewer; therefore, inter-annotator agreement was not assessed in this study. This limitation is acknowledged as an important direction for future work.

Within the validation subset manually examined, all 44 articles classified as relevant by the system were confirmed as relevant ($TP = 44$, $FP = 0$), and all 44 articles classified as non-relevant were also confirmed as non-relevant ($TN = 44$, $FN = 0$). This resulted in perfect agreement between automated and manual classification within this specific validation subset, yielding precision, recall, F1-score, and specificity values of 1.00 for this limited sample. These values describe agreement within the manually inspected sample and should not be interpreted as estimates of corpus-level screening performance.

To further assess the risk of false negatives, an extended validation step was conducted by manually inspecting an additional subset of articles from the excluded pool beyond the initial validation sample. This analysis identified two articles that were considered relevant despite being classified as non-relevant by the automated system ($FN = 2$).

Because the extended validation was conducted on a limited and non-exhaustive subset of excluded articles, these findings should be interpreted as evidence that false negatives may occur rather than as a basis for estimating corpus-level recall.

When incorporating these additional findings, the results indicate that, while the proposed methodology appears to achieve high precision within the evaluated subset, a small number of false negatives may occur in cases where semantic similarity scores fall near the decision threshold. This observation is consistent with the trade-off between precision and recall in similarity-based screening systems.

Although the performance metrics observed in the initial validation subset are high, these results should be interpreted with caution. The evaluation was conducted on a limited and partially extended sample and, therefore, may not fully generalize to the entire corpus. Nevertheless, the results suggest that the proposed approach is capable of identifying relevant studies within the evaluated context. To further contextualize these results, an estimate of statistical variability is provided in the following.

Given the limited size of the validation subset and the absence of a corpus-wide ground-truth annotation, no formal

statistical inference regarding global screening performance was performed. Consequently, the observations reported in this subsection should be interpreted as indicative evidence on the behavior of the proposed workflow within the evaluated sample.

The objective of this analysis is not to establish benchmark-level classification performance, but to provide an initial assessment of the internal consistency and practical feasibility of the proposed multi-criteria screening workflow.

D. DOMAIN SPECIFIC CHALLENGES IN ROCK ART LITERATURE

Rock art research presents several domain specific characteristics that directly affect automated semantic retrieval. First, the field is highly visual: many articles devote a substantial portion of their content to figures, multispectral plates, 3D reconstructions, and photographic datasets, resulting in low lexical density in both abstracts and full texts. This limits the amount of textual evidence available for embedding-based similarity computation.

Second, abstracts frequently emphasize the archaeological context—geographic setting, chronology, and conservation state, while providing little or no detail about digital image processing techniques. As a consequence, articles containing substantial methodological contributions often produce low or moderate abstract similarity scores.

Third, many methodological descriptions employ general purpose vocabulary (e.g., “digital enhancement,” “statistical analysis,” “image classification”) without explicitly referencing rock art until later sections. This may result in “hidden relevance” cases detectable only by full text analysis.

These characteristics explain the presence of articles that were thematically relevant but semantically weak at the abstract level, as well as the recovery of studies exclusively through full text similarity or keyword–full-text combinations. They also highlight the need for a multilevel semantic approach such as TriSem-LLM, capable of compensating for the structural heterogeneity of the domain.

E. CRITICAL ANALYSIS

1) ASSESSMENT OF CORPUS COVERAGE

A critical requirement of PRISMA-aligned methodologies is to ensure that the set of retrieved documents provides broad coverage to support a transparent and methodologically grounded synthesis of the available evidence. In the present study, the search strategy was intentionally designed to maximize breadth and recall, combining multiple large academic databases (SCOPUS, Web of Science, ScienceDirect, IEEE Xplore, Springer, Taylor & Francis, and MDPI) with multilingual queries targeting both technical and archaeological terminology.

The final dataset comprised 799 retrieved documents, of which 679 contained valid abstracts and full texts suitable for semantic analysis. This volume significantly exceeds the size of corpora typically found in domain specific systematic

reviews in digital archaeology and digital heritage, suggesting that a substantial portion of the relevant literature of the field may be represented.

The results of the screening process further support the plausibility of the adopted corpus coverage strategy. The 44 articles recommended by the TriSem-LLM pipeline cover the principal lines of research historically associated with digital processing in rock art studies, such as decorrelation stretch, PCA-based enhancement, multispectral imaging, RTI, photogrammetry (SfM), 3D laser scanning, deep learning-based motif classification, and hybrid 2D/3D documentation workflows. The manual inspection did not indicate the absence of major thematic groups within the recommended set, although this assessment is based on a qualitative inspection.

In addition, the identification of only a small number of false negatives during the extended inspection provides additional evidence regarding the behavior of the screening workflow within the inspected sample. Although absolute exhaustiveness is unattainable in any systematic review, the combination of multiple databases, multilingual coverage, and multi-criteria semantic screening provides preliminary support for the view that the collected corpus is reasonably broad to enable a transparent and representative synthesis within the scope of this study.

In summary, the corpus obtained in this study appears to meet the expected methodological standards for breadth and thematic coverage in interdisciplinary systematic reviews and provides a structured foundation for the substantive analysis developed in subsequent work.

Taken together, the breadth of the search strategy, the multilingual coverage, and the minimal false negative rate observed during manual inspection suggest that the corpus obtained is reasonably broad and suitable to support a systematic and transparent review within the scope of this study.

2) INTERPRETATION OF PRECISION AND F1 SCORE

The results of manual validation (Section V-C) indicate that all articles recommended by TriSem-LLM were assessed as relevant within the manual validation subset, while only a small proportion of excluded records were later identified as false negatives. From these results, the observed agreement within the recommended subset corresponds to a precision of 1.00 and a local F_1 score of 1.00 within this specific validation context.

These values should not be interpreted as global performance estimates, but rather as indicators of agreement between automated and human decisions within the recommended sample. The absence of false positives within the validated subset suggests that the multi-criteria strategy adopted by TriSem-LLM may favor conservative inclusion decisions, which is desirable in systematic review pipelines that prioritize transparency and interpretability.

However, the identification of a small number of false negatives in the excluded sample signals that full recall

is not guaranteed. This behaviour is expected in domains characterized by low textual density, extensive reliance on images, and heterogeneous terminology, where semantically relevant information may not be strongly expressed in abstracts or may appear only in specific segments of the full text. Consequently, the perfect F_1 score observed in the recommended subset reflects local agreement within the evaluated subset rather than complete corpus-level performance.

In summary, the observed precision and F_1 values reflect consistent agreement between automated and manual screening decisions within the evaluated subset, suggesting that the methodology is capable of identifying a coherent and thematically aligned set of articles for qualitative synthesis within the evaluated context.

3) WHY MULTILEVEL SCREENING IS STRUCTURALLY NECESSARY

The observed screening outcomes suggest that no single criterion—keywords, abstract similarity, or full text similarity—is sufficient on its own to consistently capture relevant studies in this domain. Keyword matching alone may exclude articles that introduce substantial methodological contributions without explicitly referencing technical terminology in the abstract. Similarly, abstract-based similarity may fail to identify studies whose methodological details appear primarily in later sections of the text. Although full text similarity provides broader contextual coverage, its effectiveness can be reduced in image dominated publications with low lexical density.

The multi-criteria model implemented in TriSem-LLM addresses these complementary limitations by treating each criterion as an independent source of evidence. Articles recovered through multiple criteria tend to exhibit more concentrated similarity score distributions and stronger semantic alignment with the research question, whereas articles identified through a single criterion often correspond to edge cases shaped by domain specific textual characteristics.

Taken together, these observations suggest that multilevel screening constitutes a methodologically appropriate design choice for heterogeneous, image rich domains such as digital rock art research, where terminology is inconsistent and methodological information may not be uniformly expressed in abstracts.

F. DISCUSSION AND FINAL CONSIDERATIONS

The exploratory analysis of the results suggests the practical usefulness of the implemented multi-criteria strategy, which combines lexical and semantic filters to support the initial screening process. The distribution of articles across different combinations of criteria indicates that keyword matching contributes to preliminary identification, while the inclusion of cosine similarity measures enables the recovery of records that may not be captured through lexical matching alone. These observations suggest the relevance of integrating

multiple dimensions of evidence to reduce terminological bias and broaden the coverage of documents.

The prioritization process, based on the highest similarity observed between abstract and full text representations, produces an ordered list that may help guide the subsequent critical review. This procedure is particularly valuable in contexts involving large volumes of retrieved literature, where human capacity to manually track and assess each document is constrained.

The LLM-assisted analysis may support the preliminary qualitative inspection and structured synthesis of the identified techniques and their temporal evolution. Although it does not replace detailed expert reading, this approach illustrates the feasibility of accelerating exploratory stages and evidence mapping, positioning LLMs as complementary tools within systematic review workflows rather than substitutes for human judgment.

These observations are consistent with the findings reported by Roth, who noted that despite the growing availability of machine learning tools for systematic reviews, challenges related to usability and reproducibility continue to limit their widespread adoption [9]. In this context, TriSem-LLM contributes by proposing the integration of multiple functionalities into a single workflow, offering multilingual support and explicit decision traceability aligned with PRISMA recommendations, thereby reducing fragmentation and supporting reproducible review practices.

Although the *TriSem-LLM* pipeline was developed and evaluated using a case study focused on rock art research, its core components—multilingual semantic embeddings, keyword-based heuristics and LLM-assisted synthesis—are intended to be methodologically transferable, subject to domain specific adaptation and validation. The methodology may be particularly relevant for fields characterized by heterogeneous or evolving terminology. Potential application areas include computational sustainability, digital humanities, or bio-inspired robotics, where traditional keyword-based search strategies may be challenged by non standardized vocabularies. The modular design of the pipeline allows researchers to adapt semantic models, keyword sets, and LLM prompts to the requirements of specific research questions. In addition, reliance on the PRISMA 2020 Explanation and Elaboration [4] contributes to clarity in methodological reporting and alignment with established best practices.

To contextualize the contribution of the proposed pipeline, Table 4 summarizes the key differences between TriSem-LLM and widely adopted screening systems, situating it conceptually within the landscape of existing approaches.

Table 4 suggests that, unlike existing tools and embedding-based approaches, TriSem-LLM integrates multiple screening stages within a unified and traceable pipeline.

This comparison positions TriSem-LLM as an integrated multi-criteria workflow that combines multilingual semantic analysis with explicit decision traceability. Therefore, it may

occupy a distinct methodological niche as a transparent and reproducible alternative, positioned between conventional screening platforms and generic transformer-based similarity approaches.

It is important to note that the comparison presented in Table 4 is qualitative in nature and is intended to highlight methodological differences rather than provide a direct performance benchmark. The evaluated systems differ substantially in their underlying assumptions, interaction paradigms, and evaluation protocols. As a result, direct quantitative comparison on a common dataset is non-trivial.

In particular, tools such as Rayyan and ASReview are designed to support interactive human-in-the-loop screening processes, whereas TriSem-LLM operates as a fully automated multi-stage pipeline. Similarly, standalone transformer-based approaches typically address isolated classification tasks rather than end-to-end screening workflows.

As a result, the proposed comparison focuses on structural and methodological characteristics, including semantic capabilities, full-text processing, and decision traceability. A standardized quantitative benchmarking against established datasets (e.g., CLEF TAR or SYNERGY) is an important direction for future work and would enable a more direct performance comparison across systems.

Existing systems such as Rayyan and ASReview have demonstrated substantial workload reductions under specific conditions. Rayyan has been reported to achieve average workload savings of approximately 49% in a recall of 95%, with considerable variability between datasets [8]. ASReview has also been shown to reach recall levels greater than 95% while reducing the screening effort [9], [14], although its adoption in practice remains uneven [9].

Transformer-based approaches have achieved strong performance in domain-specific tasks; for example, Du et al. reported accuracy values of up to 0.90 and F1-scores of around 0.79–0.80 for disease-focused abstract selection using models such as PubMedBERT and BioBERT [10]. However, such approaches typically focus on single-stage classification tasks and do not provide a complete, traceable screening workflow.

Within this context, TriSem-LLM contributes a complementary approach that emphasizes multilingual coverage, multi-criteria decision logic, and explicit decision traceability. In addition, the extended validation analysis, including the identification of false negatives, provides additional evidence on the ability of the proposed methodology to reduce the risk of premature exclusion while maintaining decision traceability.

Finally, several limitations identified during the application of the methodology are discussed in the following section.

G. SCOPE OF APPLICABILITY AND GENERALIZATION

Before interpreting the broader applicability of the proposed methodology, it is important to distinguish between methodological transferability and empirical generalization. The results reported in this study arise from a single, domain

TABLE 4. Comparison of TriSem-LLM with existing systematic review screening approaches.

Tool	Approach Type	Semantic	Full-text	Transparency	Pros	Cons
Rayyan	Manual-assisted screening	No	No	Low–Medium	Easy to use; speeds up title/abstract screening	Relies heavily on manual decisions; limited automation; opaque ranking logic
ASReview	Active learning ML	Partial	No	Medium	Efficient prioritization; supports high recall screening	Limited interpretability; requires iterative training and user input
Transformer-based Models (general use)	Embedding-based semantic similarity	Yes	Partial	Low	Strong semantic representation; flexible integration	Not a complete pipeline; requires external orchestration; limited decision traceability
TriSem-LLM (proposed)	Hybrid NLP + LLM pipeline	Yes	Yes	High	Multilingual screening; no early exclusion; modular pipeline; full traceability of decisions (keywords, abstract, full-text scores)	Higher computational cost; increased pipeline complexity

specific case study and should therefore be interpreted as evidence of internal coherence and feasibility within this context, rather than as a demonstration of universal performance across unrelated domains.

The contribution of TriSem-LLM primarily consists in the design and integration of a transparent, multi-criteria screening workflow that combines multilingual semantic embeddings, lexical heuristics, and LLM-assisted synthesis. Although individual components of the pipeline are based on widely adopted techniques, their configuration, parameterization, and observed behaviour are inherently influenced by domain characteristics such as terminology, publication structure, and the balance between textual and visual content.

Consequently, any extension of the methodology to other research areas should be understood as a matter of methodological transferability rather than a direct generalization of empirical results. Applying TriSem-LLM to new domains may require adaptation of keyword sets, embedding models, similarity thresholds, and prompt design to reflect domain specific conventions and reporting practices.

This clarification aims to ensure that the scope of the present evaluation is clearly defined while leaving open the potential for future applications of the methodology under appropriately adjusted conditions.

VI. LIMITATIONS AND FUTURE WORK

Despite the structured design of the *TriSem-LLM* pipeline and the encouraging observations obtained in this case study, several limitations remain and point to opportunities for further development. The LLM-assisted synthesis component supported exploratory temporal analysis and trend identification; however, its application also revealed challenges. Even with a low generation temperature (0.1), variability in focus and level of detail of the generated summaries was observed, occasionally requiring minor human revision to ensure faithful representation of the source material. In addition, the generative process remains partially opaque, making it difficult to fully trace how specific conclusions are derived.

These aspects reinforce the need for systematic human oversight and validation of all LLM-generated outputs.

Consequently, while LLMs can act as valuable aids for exploratory synthesis and evidence organization, they should not be considered substitutes for expert human judgment in this context. This perspective is consistent with recent findings that indicate that, although automation techniques have been proposed for multiple stages of the systematic review workflow, their adoption in practice remains limited and strongly dependent on human supervision [8]. Khraisha et al. similarly emphasized that LLMs should be integrated as supportive tools within transparent oversight frameworks rather than as replacements for human reviewers [15].

The choice of the *all-mpnet-base-v2* embedding model, while generic, was motivated by its documented effectiveness in scientific text similarity tasks. Zhang and Lencioni [11] reported a strong ranking behaviour for relevant documents using this model. Nevertheless, future work may explore the use of domain adapted or fine tuned embeddings trained in specialized academic corpora to further refine semantic discrimination.

In parallel, alternative screening approaches merit consideration. As highlighted by Du et al., conventional machine learning algorithms such as XGBoost can remain competitive when enriched with additional metadata features, offering potential trade-offs between semantic expressiveness and computational efficiency [10]. Future adaptations of the TriSem-LLM pipeline may explore such configurations in resource constrained environments.

The limitations of the context window of Transformer-based models were mitigated through adaptive text segmentation. Future strategies may incorporate structure aware segmentation based on document sections (e.g., IMRyD) or leverage models with extended context windows as they become more accessible.

Texts characterized by generic or high level language may be overestimated by mechanisms based on similarity. Although the fusion of abstract and full text analysis partially mitigates this effect, future work may investigate dynamic

weighting strategies or supervised classifiers to further refine screening behaviour.

Finally, OCR errors, incomplete extraction, and missing metadata can affect screening outcomes. Automated integrity checks and preprocessing validation steps will be considered in future versions of the pipeline. Future work will also explore a deeper integration of LLMs in earlier stages of the screening process, as well as the application of the methodology to additional domains. A subsequent study will focus on the thematic synthesis of the 44 selected articles.

VII. CONCLUSION

This study introduced the TriSem-LLM methodology, an automated pipeline that combines multilingual semantic embeddings, lexical heuristics, and synthesis-assisted by large language models to support systematic review of the literature in interdisciplinary contexts. The methodology was applied to a substantial corpus of scientific articles using a research question focused on digital image processing techniques in rock art studies, illustrating how semantic and lexical signals can be integrated within a transparent and traceable screening workflow.

The exploratory analysis indicated that the multi-criteria strategy adopted in TriSem-LLM may contribute to reducing the volume of articles requiring manual inspection, while preserving broad thematic coverage within the evaluated corpus. The generation of prioritized rankings based on semantic similarity between abstracts and full texts provided a structured basis for subsequent qualitative analysis. In addition, the LLM-assisted synthesis supported the exploratory organization of technical evidence in relation to the research question, complementing traditional review practices.

Manual validation performed on the retained subset of articles showed consistent agreement between automated screening decisions and human assessment within the evaluated sample. Although this validation was intentionally limited in scope, it suggests the internal coherence and traceability of the proposed workflow rather than constituting a global performance evaluation. These observations suggest that the pipeline can help reviewers manage the complexity of screening while maintaining transparency in decision making.

Overall, the findings suggest the feasibility of the TriSem-LLM approach as a reproducible and adaptable methodological framework, whose empirical behaviour should be re-evaluated when applied to new domains. Future work may investigate its application to additional domains, the incorporation of iterative update mechanisms for living reviews, and the development of complementary thematic analysis based on the selected literature, further contributing to transparent and sustainable research synthesis practices.

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CONFLICTS OF INTEREST

The authors declare no conflict of interest.

DATA AND CODE AVAILABILITY STATEMENT

All code, scripts, and supporting materials required to reproduce the proposed pipeline are publicly available in the project repository <https://github.com/SVBJardim/TriSem-LLM>. The repository includes the implementation of the full workflow, processed data, and detailed instructions for reproducing the results. Due to copyright restrictions, full-text articles obtained from subscription-based databases cannot be redistributed; however, metadata (e.g., titles, abstracts, and identifiers) and instructions for data retrieval are provided to ensure reproducibility.

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TASSIA CARVALHO received the degree in information systems. She is currently pursuing the master's degree in computer engineering with the University of Beira Interior (UBI), Portugal. She has prior experience as a Lecturer in information technology in Brazil. She completed a Lato Sensu Postgraduate Programme with UBI. She is also conducting applied research within the RA3I project, focusing on the design of automated methodologies for systematic literature reviews, combining large language models (LLMs), semantic embeddings, and document screening techniques. Her research interests include natural language processing, scientific decision-support systems, and the application of artificial intelligence in interdisciplinary contexts.

SARA GARCÊS received the degree in archaeology from Minho University, in 2007, the M.Sc. (master's) degree in prehistoric archaeology and rock art from the Polytechnic University of Tomar Erasmus Mundus Program IMQP, in 2009, and the Ph.D. degree in quaternary, materials and cultures from the University of Trás-os-Montes e Alto Douro, in 2017. She is currently an Adjunct Professor with the Polytechnic University of Tomar. She is also an Integrated Researcher and the Vice-Director of the Geosciences Center and the Coordinator of the Rock Art Archaeology Laboratory, Instituto Terra e Memória, participating in the rock art and prehistoric archaeology research lines. Her main research interests include prehistoric rock art, dating and archaeometry, and symbolic systems of hunter-gatherer societies in the Tagus Valley and the Iberian Peninsula. She has authored or co-authored several scientific articles in indexed journals, including the *Journal of Archaeological Science*, *L'Anthropologie*, *Rock Art Research*, and *Heritage*; and edited the volume *The Prehistoric Rock Art of Portugal: Symbolising Animals and Things*. She has coordinated and participated in national and international research projects funded by FCT and European programs, including those related to the documentation, conservation, and valorization of rock art heritage in Portugal and Spain. She regularly collaborates with research teams in Spain, U.K., Colombia, Türkiye, South Africa, and Brazil; and serves as a peer reviewer for scientific journals in the fields of archaeology and heritage sciences.

SANDRA JARDIM received the degree in electronics and telecommunications engineering from the University of Aveiro, in 1992, and the M.Sc. degree in electrical and computer engineering and the Ph.D. degree from the Instituto Superior Técnico, University of Lisbon, in 2002 and 2008, respectively. She is currently a Coordinate Professor with the Polytechnic Institute of Tomar, where she works as the Director of the Bachelor's Degree in Informatics and Multimedia Technologies, the Master's Degree in Analytics and Business Intelligence, and the Bachelor's Degree in Computing and Logistics. She also chairs the Technical-Scientific Council of the Abrantes School of Technology and also an Integrated Researcher at the Smart Cities Research Center. She has participated in and coordinated several funded applied research projects, including RA3I and DABB, focusing on machine learning, computer vision, and digital image processing. She is the author or co-author of multiple scientific articles and conference papers and an active Reviewer for several indexed journals, such as IEEE Access, *PLOS One*, EAAI (Elsevier), *Healthcare Analytics*, and the *Journal of Imaging*. She is a Guest Editor of the Special Issue *Pattern Recognition and Image Processing: Latest Advances and Prospects* of the *Electronics* journal and a member of the Editorial Board of *American Journal of Computer Science and Technology*. Her main contributions are in the fields of machine learning, deep learning, computer vision, and image processing.

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